DECK 10 · CORPORATE ISSUERS

Working Capital, Capital Allocation & Business Models

Short-term cycle, long-term investment, and the business model underneath both

CFA Level I · Vol 3 Learning Modules 4, 5, 7

Roadmap — Working Capital, Capital Allocation & Business Models

3 sections · ~2 hours · Built bottom-up from CFA readings

01

Working capital & liquidity

CCC mechanics · Inditex & discount retailers · supplier financing · liquidity ratios · drags & pulls · conservative vs aggressive

02

Capital investments & allocation

project types · 4-step process · NPV / IRR / ROIC · inflation · cognitive & behavioral pitfalls · real options with decision tree

03

Business models

the five questions · eight conventional archetypes · twelve pricing models · platforms & network effects · B2B vs B2C · unit economics (CAC/LTV)

Learning objectives

By the end of this deck you should be able to:

1. Compute a cash conversion cycle from activity ratios, interpret negative CCCs, and price supplier trade credit against a bank loan.

2. Evaluate firm liquidity using current / quick / cash ratios and distinguish primary from secondary liquidity sources, including drags and pulls.

3. Apply NPV, IRR and ROIC to capital-investment decisions, apply the three allocation principles, and name the three cognitive errors and four behavioral biases the curriculum identifies.

4. Identify the four types of real option (timing, sizing, flexibility, fundamental) and compute a project NPV that embeds a decision tree.

5. Decompose a business model into its core questions, classify it into one of eight conventional archetypes, and compute the unit economics of a subscription business (CAC, LTV, payback).

SECTION 01

Working capital & liquidity

The operating cycle turns materials into cash; the cash conversion cycle is the gap the firm has to finance after suppliers have funded what they will

Inventory and receivables lengthen the cycle; payables shorten it. A negative CCC funds the firm.

$$CCC = DOH + DSO - DPO$$

where: DOH = days inventory on hand · DSO = days sales outstanding · DPO = days payable outstanding.

Industry (2020 S&P 1500 average)	CCC (days)	What drives it
Pharmaceuticals	180	Regulatory stockpiles, long DOH
Apparel	123	Seasonal inventory build
Semiconductors	121	Long multi-stage production
All S&P 1500 companies	78	The 2020 benchmark
Auto & auto parts	66	Just-in-time parts inventory
Retail	61	Card sales at the till, tiny DSO
Software	22	Subscriptions billed in advance
Airlines	lowest	Prepaid tickets, no goods inventory

Source: CFA Institute, Curriculum Vol 3, Learning Module 4, §2 "Cash Conversion Cycle" and Exhibits 1, 4 and 5, pp. 101–103.

Inditex runs a structurally negative cash conversion cycle — its customers pay 86 days before its suppliers do

Fast fashion, payment at the till, a vast supplier base. Watch the ratio bases in the right-hand column.

Inditex (EUR millions, or days)	2021	2020	Ratio base
Accounts receivable	842	715	–
Inventories	3,042	2,321	–
Accounts payable	6,199	4,659	–
Sales (revenue)	27,716	20,402	–
Cost of goods sold	11,902	9,013	–
Days sales outstanding (DSO)	11 d	13 d	AR / Sales × 365
Days of inventory on hand (DOH)	93 d	94 d	Inv / COGS × 365
Days payable outstanding (DPO)	190 d	189 d	AP / COGS × 365
CCC = DOH + DSO – DPO	**-86 d**	**-82 d**	

AND THE WORKING CAPITAL SIGN FOLLOWS THE CYCLE

Net working capital 2021 = 842 + 3,042 − 6,199 = −EUR 2,315m, or −8.4% of sales (2020: −8.0%). Operating accounts only — cash, securities and short-term debt are excluded. A short cycle means low working capital to sales; here it is negative, so growth releases cash instead of consuming it.

Source: CFA Institute, Curriculum Vol 3, Learning Module 4, Inditex case study, pp. 104 and 108.

Five US discount retailers, one formula, a 28-day spread — and DSO explains none of it

All five take cash or card at the register, so DSO is 2–7 days for everyone. Inventory and payables do the work.

Retailer (2021)	DOH	DSO	DPO	CCC = DOH + DSO − DPO
Walmart	48 d	5 d	47 d	48 + 5 − 47 = +6 d
Target	68 d	2 d	75 d	68 + 2 − 75 = −5 d (shortest)
Costco	29 d	3 d	34 d	29 + 3 − 34 = −2 d
TJX	63 d	7 d	47 d	63 + 7 − 47 = +23 d (longest)
Ross Stores	61 d	2 d	63 d	61 + 2 − 63 = 0 d

WHERE THE SPREAD ACTUALLY COMES FROM

Costco sells roughly 54% food and food-related items: perishables must move before they spoil, so DOH is only 29 days. Target carries the deepest inventory of the five at 68 days, but also extracts the most supplier financing at 75 days payable — and that is what flips it negative. TJX is Target's mirror image: 63 days of inventory funded by only 47 days of payables, so it pays for the stock long before it sells it.

Source: CFA Institute, Curriculum Vol 3, Learning Module 4, Example 2 "Cash Conversion for US Discount Retailers", pp. 106–107.

Forgoing a 2/10 net 30 discount is borrowing from your supplier at 44.6% — a bank at anything below that is strictly cheaper

You pay the discount to use the supplier's money for (net days − discount days). Annualise it before you compare.

$$EAR = \left[1 + Disc\% / (100\% - Disc\%) \right]^{(365 / (net\ days - disc\ days))} - 1$$

where: Disc% = prompt-payment discount · net days = full payment period · disc days = discount period.

KEOWN CORP — 2/10 NET 30 vs BANK

Supplier: 2% off if paid within 10 days, otherwise the full amount at day 30. Keown lacks the cash to pay on day 10.

1. Interest cost = 2% / (100% − 2%) = 0.02041
2. Periods per year = 365 / (30 − 10) = 18.25
3. EAR = (1.02041)^18.25 − 1 = 44.6%
4. Borrow at 7.7%, pay day 10, take the discount.

SUPPLIER 44.6% vs BANK 7.7% → USE THE BANK

Trade credit terms	Implied EAR
1/10, net 50	9.6%
1/14, net 30	25.8%
3/15, net 60	28.0%
2/15, net 40	34.3%
2/10, net 30	44.6%

[1] Cost rises with the discount given up and falls with the extra days bought: 1/10 net 50 buys 40 extra days for 1% — the cheapest of the five.

Source: CFA Institute, Curriculum Vol 3, Learning Module 4, Example 1 "Keown Corporation" and Question Set Q4–Q5, pp. 105 and 110–111.

Licht Vernieuwend's current ratio improved and its liquidity got worse — because the improvement was inventory

One denominator, three numerators. The order is always Current ≥ Quick ≥ Cash — only the tightness changes.

CURRENT RATIO

Current assets / current liabilities.

Broadest. Above 1 means positive total working capital — nothing more.

QUICK RATIO

(Cash + ST securities + receivables) / current liabilities.

Drops inventory. Above 1 = can pay without selling stock, if every receivable collects.

CASH RATIO

(Cash + ST securities) / current liabilities.

Most conservative. At or above 1 = settle everything today, collecting nothing and selling nothing.

Licht Vernieuwend N.V.	20X1	20X2	Change
Cash ratio = (Cash + Sec) / CL	0.60	0.50	−0.10 worse
Quick ratio = (Cash + Sec + AR) / CL	1.00	0.89	−0.11 worse
Current ratio = CA / CL	1.36	1.52	+0.16 better
Net working capital (EUR m)	16	65	+49 tied up
Current ratio rose only because inventory doubled	**66**	**130**	**Less liquid**

Source: CFA Institute, Curriculum Vol 3, Learning Module 4, §3 "Liquidity", Exhibit 9 and Knowledge Check, pp. 117–120.

Drags slow the cash coming in, pulls speed up the cash going out — and reaching for a secondary source is itself the signal

Primary sources are the ordinary course. Secondary sources raise cash at a cost — using them is news in itself.

DRAGS — INFLOWS LAG

· Uncollected receivables — the longer they age, the lower the odds of collection
· Obsolete inventory — held too long, then sells only at a discount
· Borrowing constraints — credit tightens, short-term debt gets dearer or vanishes

PULLS — OUTFLOWS ACCELERATE

· Paying early when no discount is on offer
· Suppliers cutting your credit limit after late payments
· Banks capping short-term lines
· A chronically thin liquidity position, made worse by aggressive policy

Liquidity source	Curriculum examples	Signal
Primary — cash & securities	Bank balances, T-bills, commercial paper held	Normal
Primary — borrowings	Bank lines, bond or CP issuance, supplier credit	Normal
Primary — cash flow from business	The long-run engine; read the cash flow statement	Normal
Secondary — cut dividend or capex	Delta deferred USD 500m of capex, 2020	Stress
Secondary — equity or renegotiate	Singapore Airlines SGD 8.8bn rights issue, 2020	Stress
Secondary — sell assets, file Ch.11	Disposals at a discount; insolvency reorganisation	Distress

Source: CFA Institute, Curriculum Vol 3, Learning Module 4, §3 primary and secondary liquidity sources, drags and pulls, pp. 113–117.

Working-capital policy is a spectrum, and the choice is really a single trade: financing cost against rollover risk

Conservative buys safety with cost. Aggressive buys cost with risk. Moderate matches the funding to the asset.

Dimension	Conservative	Moderate ("matched")	Aggressive
Current assets vs sales	Larger	Moderate	Smaller
Funding mix	More long-term + equity	Match asset to funding	More short-term debt
Financing cost	Highest	Middle	Lowest
Rollover risk	Minimal	Variable share only	High — bites in a crisis
Flexibility under stress	High	Medium	Low
Rate view that suits it	Flat to rising	Mixed	Stable to falling
Typical user	Early-growth firms	Mature, diversified	Low-margin, predictable

WHEN AGGRESSIVE BITES

Rollover risk is the whole story. In 2008–09 and again in the first phase of COVID in 2020, firms that had funded permanent current assets with short-term paper found the market shut and raised equity at distressed prices instead. Match-funding is the defensive default for any business with a cyclical revenue line.

Source: CFA Institute, Curriculum Vol 3, Learning Module 4, §4 Exhibits 10–13 and the stated rationales, pp. 123–127.

SECTION 02

Capital investments & allocation

Four categories of capital investment — and only one of them can rationally proceed with a negative NPV

Two maintain the business, two grow it. Regulatory projects are the only non-discretionary ones.

Category	Purpose	Risk	Curriculum example
Going concern (maintenance)	Replace ageing assets; cut cost	Low	Data-centre cooling upgrade
Regulatory / compliance	Meet rules, standards; avoid fines	Low	Danske Bank AML remediation
Expansion of existing business	Scale up or widen scope; core R&D	Low–med	Sony buys Bungie, USD 3.6bn
New lines of business & other	Enter an unrelated industry	High	Kirin buys into Fancl, ¥129bn

HOW ANALYSTS SPLIT THE CAPEX LINE

Issuers do not disclose the split. Analysts proxy maintenance capex with D&A expense — most accurate for short-lived assets, where historical cost is close to replacement cost.

Expansion capex ≈ Total capex − D&A.

MATCH FUNDING

Match the debt tenor to the asset life. A utility replacing 30-year turbines issues a 30-year bond.

Borrow shorter than the asset and you carry rollover risk; borrow longer than you need and you pay a higher rate or buy the debt back.

Source: CFA Institute, Curriculum Vol 3, Learning Module 5, §2 Exhibit 2 and Examples 1–3 (Danske, Sony, Kirin), pp. 139–145.

Capital allocation is a four-step loop, and NPV and IRR only appear in step two — the value is created in steps three and four

A positive NPV is necessary, not sufficient. Step 3 decides what actually gets funded, step 4 decides what you learn.

1

1. Idea generation

Managers, business-unit heads and external consultants propose projects. Most ideas come from inside existing operations — which is exactly where the inertia bias enters.

2

2. Investment analysis

Forecast the amount, timing, duration and volatility of after-tax cash flows. Compute NPV and IRR. Use a required return set by the project's risk — never by how it is financed.

3

3. Planning & prioritisation

Rank on risk-adjusted value across the whole portfolio, under financing, strategic and ESG constraints. When value-creating projects run out, return the capital to shareholders.

4

4. Monitoring & post-audit

Compare actuals with projections to expose systematic optimism, enforce operating discipline, and generate the next round of ideas. Exercise real options as information arrives.

Source: CFA Institute, Curriculum Vol 3, Learning Module 5, §3 Exhibit 3 "Steps in the Capital Allocation Process", pp. 146–147.

NPV converts a project into one number in currency — the amount shareholder wealth rises today if the forecast holds

NPV is in euros, so it can be added up and compared. IRR is a rate, which is why it cannot rank projects.

$$NPV = CF_0 + \Sigma\ CF_t / (1 + r)^t \qquad \text{Invest if } NPV \geq 0$$

where: CF_0 = initial outlay (negative); CF_t = after-tax cash flow at t; r = return investors could earn on a similarly risky investment.

GERHARDT CORPORATION — EUR 50M TODAY, r = 10%

After-tax cash flows of EUR 16m a year for four years, then EUR 20m in year 5. All amounts in EUR millions.

1. PV(y1) = 16 / 1.10 = 14.545 PV(y2) = $16 / 1.10^2$ = 13.223
2. PV(y3) = $16 / 1.10^3$ = 12.021 PV(y4) = $16 / 1.10^4$ = 10.928
3. PV(y5) = $20 / 1.10^5$ = 12.418 Sum of inflows = 63.136
4. NPV = −50 + 63.136 = +13.136 ⇒ pay 50 today for something worth 63.136

NPV = +EUR 13.136M → INVEST (+26% on the outlay, in present value)

[1] Excel: the NPV function starts at t = 1, so the outlay sits outside it — =NPV(0.10,16,16,16,16,20) − 50 = 13.136.
[2] NPV ≥ 0 is necessary but not sufficient: competing projects, financing limits and strategic fit still bind in step 3.

Source: CFA Institute, Curriculum Vol 3, Learning Module 5, §3 Equation 1 and Example 4 "Gerhardt Corporation NPV", pp. 147–149.

Inflation is a matching problem: nominal cash flows at a nominal rate, real cash flows at a real rate — matched, they give the same answer

Either convention is fine. Mixing them is the cognitive error — and it silently inflates or deflates the NPV.

$$(1 + r\_nominal) = (1 + r\_real) \times (1 + \pi) \Rightarrow r\_real = (1 + r\_nominal) / (1 + \pi) - 1$$

where: π = expected inflation. Mixing conventions — nominal CFs at a real rate, or the reverse — is meaningless.

Illustrative project, π = 5%	t = 0	t = 1	t = 2	t = 3	NPV
Real cash flow (today's prices)	−100.0	40.0	40.0	40.0	–
Nominal CF = real CF × 1.05^t	−100.0	42.0	44.1	46.3	–
Nominal CF discounted at 10.00%	–	38.18	36.45	34.79	+9.42
Real CF discounted at 4.76%	–	38.18	36.45	34.79	+9.42
Matched conventions → same answer					**+9.42**

[1] Consistency is not immunity: Tolonen matched real flows to a real rate, lithium then rose 10% against a 5% forecast, and the project still moved.

Source: CFA Institute, Curriculum Vol 3, Learning Module 5, §4 cognitive error "Inconsistent treatment of inflation" and the Tolonen question, pp. 160–162. Figures below are illustrative.

IRR is the rate that zeros NPV — a fine accept/reject test for one project, and the wrong tool the moment you have to choose between two

Same Gerhardt cash flows: IRR = 19.52% against a 10% hurdle, so invest. Now three ways the rule falls over.

$$\text{IRR} = \text{the rate } r^* \text{ for which } \Sigma_t \; CF_t / (1 + r^*)^t = 0$$

$$\text{Invest if IRR} \geq r$$

FAILURE 1 · ROOTS

Signs flip more than once. Cash flows of −1,000 / +5,000 / −6,000 set NPV = 0 at BOTH 100% and 200%.

Excel and calculators return the lowest root and say nothing about the other.

When signs change twice, use NPV.

FAILURE 2 · SCALE

At r = 10%, one year:

A — invest 100, receive 150.
IRR 50%, NPV +36.

B — invest 1,000, receive 1,300.
IRR 30%, NPV +182.

Mutually exclusive? Take B. IRR ranks rates; only NPV ranks money.

FAILURE 3 · REINVEST

IRR only equals your realised return if every interim cash flow is reinvested at the IRR itself.

A 30% IRR quietly assumes you can redeploy at 30%. Reinvest below it and you earn less than the IRR.

NPV assumes reinvestment at r.

[1] Reinvestment: IRR assumes interim cash flows earn the IRR; NPV assumes they earn r, which the curriculum calls the more economically realistic assumption.
[2] For mutually exclusive projects, the higher NPV always wins — the curriculum is explicit that NPV is the theoretically sound criterion.

Source: CFA Institute, Curriculum Vol 3, Learning Module 5, §3 Equation 2, Example 6, the multiple-IRR case and "Which to use: NPV or IRR?", pp. 150–153.

ROIC is the capital-efficiency test an outsider can actually compute — and it splits cleanly into margin times turnover

NPV and IRR need project data you will never see. ROIC comes off the published statements — and compares directly with the required return.

ROIC = After-tax operating profit / Average invested capital = Margin × Turnover

where: Invested capital = average of beginning and ending long-term liabilities plus equity. Working capital is excluded.

CURRICULUM EXAMPLE 8 — ROIC IN YEAR 2

After-tax operating profit 24,395. Long-term debt + share capital + retained earnings: Y1 276,387; Y2 282,280.

1. Average invested capital = (276,387 + 282,280) / 2 = 279,333
2. ROIC = 24,395 / 279,333 = 8.73%
3. Investors' required return = 10%
4. 8.73% < 10% — profitable, but not earning its cost of capital

ROIC 8.73% < 10% REQUIRED → LOOK ELSEWHERE

TWO ROUTES TO THE SAME ROIC

ROIC = (After-tax profit / Sales) × (Sales / Invested capital)

· Premium route — fat margin, slow turnover. Luxury goods, patented pharma.
· Volume route — thin margin, fast turnover. Discount retail, distribution.

A high-margin firm can still earn a poor ROIC if turnover is low, and a low-margin firm can earn an excellent one if turnover is high.

[1] ROIC is an accounting measure, backward-looking, and there is no consensus on the denominator — read the trend and the peer set, not one year.

Source: CFA Institute, Curriculum Vol 3, Learning Module 5, §3 Equation 3, Example 8 and the ROIC decomposition, pp. 153–156.

Three principles prevent most of it; what remains is three cognitive errors you fix with process and four behavioral biases you fix with governance

Cognitive errors are calculation and process failures. Behavioral biases are judgement failures — those need the board.

THE THREE PRINCIPLES — APPLY THESE AND MOST PITFALLS NEVER ARISE

1 — Decide on after-tax cash flows, not accounting profit. 2 — Count only incremental cash flows: ignore sunk costs, but include cannibalisation of existing products and cost savings elsewhere in the firm. 3 — Get the timing right: shifting cash flows between periods changes both NPV and IRR.

COGNITIVE ERRORS · PROCESS

1 · Internal forecasting errors — overhead understated, competitor response ignored, wrong cost or return inputs.

2 · Ignoring the cost of internal financing — retained cash is equity withheld from shareholders, so it carries their opportunity cost. Same r whatever the source.

3 · Inconsistent treatment of inflation — nominal cash flows at a real rate, or the reverse.

BEHAVIORAL BIASES · GOVERNANCE

4 · Inertia — capex anchored to last year. McKinsey found 0.92 year-on-year correlation across 1,600+ US listed firms.

5 · Deciding on EPS or RoE — a high-NPV project can dent near-term earnings while buybacks flatter them.

6 · Pet projects — waved through without analysis, or run on flattering assumptions.

7 · No alternatives, no scenarios — no break-even, no simulation, often no history of ever having failed.

[1] Inertia evidence: Hall, Lovallo & Musters, "How to Put Your Money Where Your Strategy Is", McKinsey Quarterly, March 2012.
[2] Detection: plot segment capex against segment ROIC. Rising capex into falling ROIC is the inertia signature.

Source: CFA Institute, Curriculum Vol 3, Learning Module 5, §4 "Capital Allocation Principles and Pitfalls", pp. 159–162.

Real options are the right, not the obligation, to change course later — timing, sizing, flexibility and fundamental

Option cost is what you pay upfront to keep the choice open. Option value is what the choice turns out to be worth.

$$\text{Project NPV} = \text{NPV(without options)} - \text{Option cost} + \text{Option value}$$

where: Option value above option cost means the flexibility pays. Price it with a decision tree or an option-pricing model.

TIMING OPTION

Invest later rather than now. Give up near-term cash flow to buy information — wait for the subsidy law before building the solar farm.

SIZING OPTION

Abandon if results disappoint, expand if they are strong. This is the option embedded in any staged capital programme.

FLEXIBILITY OPTION

Price-setting when demand outruns capacity; production flexibility — overtime, extra shifts — when demand misses the forecast.

FUNDAMENTAL OPTION

The whole project hangs on an outside price. Drill the well only if oil is high enough; continue the R&D only if Phase II reads out.

BACKING OUT THE OPTION VALUE — CURRICULUM QUESTION SET

A distribution-centre upgrade is worth USD 8.9m today, or USD 9.7m if the firm waits for a renewable-energy subsidy law. Waiting costs USD 1.8m. Then 9.7 = 8.9 − 1.8 + option value, so the option value is USD 2.6m — comfortably more than the 1.8m it cost to keep the choice open.

Source: CFA Institute, Curriculum Vol 3, Learning Module 5, §5 "Real Options", Equation 4 and Question Set Q3, pp. 163–165 and 168–169.

The decision tree flips Gerhardt from reject to accept once the facility's alternative uses are priced in — the abandon branch itself is worth nothing

EUR 500m outlay. A 60% chance of 750 at t = 1. On failure, three branches at t = 2. Required return 10%.

NPV WITHOUT OPTIONS

Failure recovers nothing.

NPV = −500 + (0.6 × 750) / 1.10
= −500 + 409.09 = −90.91

→ REJECT

NPV WITH REAL OPTIONS

On failure, act again at t = 2.

NPV = −90.91 + (72 + 48) / 1.10^2
= −90.91 + 99.17 = +8.26

→ ACCEPT

Branch	Joint probability	Cash flow (EUR m)	Timing	PV contribution
Upfront outlay	100%	−500	t = 0	−500.00
Product launch succeeds	60%	+750	t = 1	+409.09
Failure → alternative product launch	0.4 × 0.3 = 12%	+600	t = 2	+59.50
Failure → sell facility to another firm	0.4 × 0.3 = 12%	+400	t = 2	+39.67
Failure → abandon the facility	0.4 × 0.4 = 16%	+0	t = 2	+0.00
NPV with real options				**+8.26**

Source: CFA Institute, Curriculum Vol 3, Learning Module 5, Example 9 "Capital Allocation Using a Decision Tree", pp. 165–166.

SECTION 03

Business models

A business model answers who, what, where, how much and how — and the question management leaves out is usually the finding

Analysts should never rely on management's own description alone — test it against what the firm actually does.

Question	What it asks	Tesla	Netflix
WHO	Target customers and segments	Aspirational middle/upper class, moving mass-market	Global mass-market households
WHAT	Product, service, value proposition	Premium EVs, storage, self-driving software	On-demand video plus originals
WHERE	Channels: how it reaches customers	Direct sales, own website, Supercharger network	Apps on every connected device
HOW MUCH	Price vs competitors, and why	Premium, justified on total cost of ownership	Tiered subscription plus an ad tier
HOW	Assets, partners, value chain	Vertical integration: batteries and software in-house	Licensed content plus originals; runs on AWS

WHY THE "WHO" GETS TESTED FIRST

Segment by geography, demographics, life-cycle stage, behaviour or wealth — the segment sets the addressable market and every forecast downstream of it. In the curriculum's Luggo case the CEO never names a customer. That is the finding.

[1] The module overview names four parts — who, what, where, how much. The lesson adds the "how": key assets, partners, suppliers and value chain.

Source: CFA Institute, Curriculum Vol 3, Learning Module 7, §2 "Defining the Business Model", Examples 1, 2 and 6, pp. 216–226.

Eight conventional archetypes cover most of the market — for these firms success comes from execution, not model innovation

Before calling any firm disruptive, work out which combination of these eight it is actually running.

Archetype	Customer	Product	Pricing	Example
Natural-resource producer	Refiners, traders	Usable raw material	Spot / forward price	Total Energies
Manufacturer	Distributors, OEM	Finished goods	Price per unit sold	L'Oréal
Distributor	Retailers	Transport, storage	Spread + service fee	McKesson
Retailer	End-users	Goods + experience	Mark-up + member fee	JD.com
Broker / marketplace	Buyers + sellers	The connection	Commission, listing	Pinduoduo
Bank	Borrowers	Loans and leases	Net interest spread	HSBC
Service producer	Businesses	Services	Service fees	Infosys
Software	Businesses	Software	Subscription fee	Shopify

THE FOUR VARIATIONS THE CURRICULUM NAMES

Private-label and contract manufacturers (Apple and NVIDIA design and market; partners build) · value-added resellers · licensing for a royalty · franchising. All recombinations of the eight.

Source: CFA Institute, Curriculum Vol 3, Learning Module 7, §3 Exhibit 5 "Conventional Business Models" and Business Model Variations, pp. 230–232.

The curriculum names twelve pricing models — each is a different theory of who is willing to pay what, and why

Price discrimination captures more of the demand curve. The only question is which axis you segment on.

TIERED
Different prices by volume or by feature tier — base vs premium vehicle trims.

DYNAMIC
Price moves with supply, demand and time. Hotel seasonality; Uber and Lyft surge.

VALUE-BASED
Price set on value delivered. A stroke drug priced off the hospitalisation it avoids.

AUCTION
Bidding discovers the price. Google and Baidu search ads; eBay merchandise.

BUNDLING
Complements sold together. Phone + cable + internet; software plus cloud storage.

RAZOR/BLADE
Cheap device, rich consumable. Razors, printer ink, consoles, lab reagents.

ADD-ON
Optional extras sold to a captive buyer after the main decision — in-app content.

PENETRATION
Discount hard to buy scale. Subsidised handsets; Amazon Echo. Regulators may object.

FREEMIUM
Free tier, paid upgrade. Used where network effects reward a large user base.

HIDDEN REVENUE
Free to the user, paid by someone else — ad-funded media; sellers pay, buyers do not.

SUBSCRIPTION
Recurring fee for access. Fixed, tiered, or metered on usage — SaaS and cloud.

LICENCE/LEASE
Access instead of ownership: leasing assets, licensing IP, franchising a whole model.

Source: CFA Institute, Curriculum Vol 3, Learning Module 7, §2 "Pricing and Revenue Models", digital pricing and recurring-revenue models, pp. 222–224.

Network effects make the user base the asset — and past critical mass they become the barrier to entry

The metric that matters here is the number of users and the rate they are joining — not this quarter's margin.

ONE-SIDED NETWORK

One type of user, and every extra user is valuable to all the others.

Telephone service; Venmo peer-to-peer payments; messaging.

Once the network is large enough, the marginal cost of one more user sits below the revenue that user brings.

MULTI-SIDED NETWORK

Two or more distinct user types, with cross-side effects.

Visa and UnionPay = cardholders × merchants. Airbnb = guests × hosts. Food delivery = consumers × restaurants × drivers.

More users pull in more merchants, which pull in more users — and it compounds.

Platform	User types	Where the barrier comes from
WeChat (Tencent)	Messengers × merchants × developers	Super-app lock-in; switching cost
Uber / Lyft	Riders × drivers	Local density and ETA
Airbnb	Guests × hosts	Listing breadth, review history
Visa / Mastercard / UnionPay	Cardholders × merchants	Universal acceptance, issuers
TripAdvisor / Yelp	Reviewers × merchants × readers	Crowdsourced review corpus

Source: CFA Institute, Curriculum Vol 3, Learning Module 7, §3 "Network Effects and Platform Business Models", Exhibit 7, pp. 233–234.

Netflix rebuilt its own model three times; the hotel industry was rebuilt around its incumbents six times — same forces, different outcome

The recurring pattern: a new channel, then network effects, then crowdsourced supply or content.

NETFLIX · THREE MODELS, ONE COMPANY

DVD rental by mail, from 1998
A website plus the postal service, when only 2% of US households owned a DVD player
Then a flat monthly fee for several at once

Streaming subscription
Same subscription logic, near-zero marginal cost per extra title; DVDs de-emphasised

Original content producer
Exclusive titles differentiate a catalogue that licensing alone had commoditised

Blockbuster took years to copy the first model.
It filed for bankruptcy in 2010 and closed in 2014.

Each wave demanded a different capex profile.

HOTEL & TRAVEL · SIX LAYERS OF CHANGE

Scale — global chains and brands
IHG: 15+ brands, roughly 6,000 locations

Franchising — Hilton largely franchises rather than owns; franchisees pay the fees

Functional separation — brand split from bricks
Host Hotels, the largest hotel REIT, is the biggest third-party owner of Marriott and Hyatt

Fractional ownership — Wyndham time-shares

OTAs — Booking, Expedia, Ctrip aggregate price and disintermediate the travel agent

Home sharing — Airbnb and Vrbo supply unique, crowdsourced accommodation

Response: direct booking, loyalty (Holiday Inn and Marriott launched theirs in 1983), soft brands.

Source: CFA Institute, Curriculum Vol 3, Learning Module 7, Example 3 "Business Model Evolution: Netflix" and Example 7 "Hotel and Travel Industry", pp. 219 and 235–236.

B2B and B2C are not just different buyers — the buyer type propagates through channel, cycle and service

The curriculum's anchor: business buyers have straightforward needs and want a return. Consumer needs can be impossible to quantify.

Dimension	B2B	B2C
Who is buying	Few, sophisticated, profit-driven	Many, diverse, part emotional
What they need	Straightforward — ship it, run it	Hard to quantify; brand and style
Sales cycle	Months: RFP, pilot, committee	Minutes to days
Main channel	Direct sales force, resellers	Retail, e-commerce, omnichannel
Ticket size	Large; few customers to reach	Small; reach is the whole game
After the sale	High-touch, SLA, integration	Self-serve, FAQ, chat
Illustrative firms	Infosys, Siemens, Salesforce	L'Oréal, Nike, IKEA, Netflix

HYBRIDS, AND WHY THEY NEED SPLITTING

Plenty of firms run both motions at once — an enterprise contract and a self-serve individual plan on the same product. Split the unit economics before comparing them with anything. The curriculum's DiaSera case is the warning: premium service promised at near-parity pricing only works if the volume actually arrives, and if it does not, management has to raise prices or abandon the promise.

Source: CFA Institute, Curriculum Vol 3, Learning Module 7, §2 on customer type and channel strategy, pp. 217–221; the remaining rows are practitioner framing.

Subscription economics reduce to three numbers: CAC, LTV, and payback — and the ratio LTV/CAC tells you if the business is viable

CAC is cash out to win a customer. LTV is the gross profit they return. Payback is how long you wait for it.

$$\text{LTV} = \text{ARPU} \times \text{Gross margin} / \text{Churn} \qquad \text{Payback} = \text{CAC} / (\text{ARPU} \times \text{Gross margin})$$

where: ARPU = average revenue per user per period · churn = share of customers lost per period.

CAC · ACQUISITION

Sales and marketing spend in the period, divided by new paid customers added.

Blended CAC mixes organic with paid. Paid CAC is the marginal number that decides whether to spend the next euro.

LTV · LIFETIME VALUE

Gross profit expected over a customer's life.

Simple form: ARPU × gross margin ÷ churn.

Better form discounts it at the cost of capital and allows for expansion revenue as accounts grow.

PAYBACK PERIOD

Months to earn the CAC back at the gross-profit run rate.

Under 12 months — strong
12 to 24 months — normal
Over 24 months — fragile

Short payback means less capital tied up and faster compounding.

[1] Practitioner rule of thumb: LTV/CAC ≥ 3× for venture-scale SaaS; payback ≤ 12 months for top-quartile public SaaS (Bessemer "Good, Better, Best").

Source: CFA Institute, Curriculum Vol 3, Learning Module 7, §2 "unit economics" framing (p. 227) + industry-standard SaaS practice.

USD 300 CAC against USD 40 ARPU at 70% margin gives a 10.7-month payback and 3.1× LTV/CAC — and churn moves only one of those two

Churn never touches payback — CAC and gross profit set that. Churn only sets how long they keep paying.

MONTHLYSTREAMER INC. — STEP BY STEP

10,000 subscribers, USD 40 ARPU a month, 70% gross margin, 3% monthly churn, USD 300 blended CAC.

1. Monthly gross profit per customer = 40 × 0.70 = USD 28 (so the book earns 10,000 × 28 = USD 280k a month)
2. Average customer life = 1 / churn = 1 / 0.03 = 33.3 months
3. LTV = 28 × 33.3 = ARPU × GM / churn = 40 × 0.70 / 0.03 = USD 933
4. Payback = CAC / monthly gross profit = 300 / 28 = 10.7 months
5. LTV / CAC = 933 / 300 = 3.1× — clears the 3× bar, but only just

LTV USD 933 · PAYBACK 10.7 MO · LTV/CAC 3.1× → VIABLE

Monthly churn	Average life	LTV (USD)	Payback (mo)	LTV/CAC	Verdict
1% best-in-class	100 mo	2,800	10.7	9.3×	Excellent
3% illustrative	33.3 mo	933	10.7	3.1×	Viable
5% weak	20.0 mo	560	10.7	1.9×	Marginal
8% severe	12.5 mo	350	10.7	1.2×	Burns cash

Source: Illustrative unit-economics example built on the "unit economics" concept in CFA Vol 3 LM 7 §2 (p. 227); the metrics themselves are industry practitioner standards, not curriculum.

KEY TAKEAWAYS

Deck 10 — Working Capital, Capital Allocation & Business Models

- CCC = DOH + DSO − DPO, and DPO carries the minus sign: a longer payable period SHORTENS the cycle. Inditex −86 days, Target −5, TJX +23.
- Get the ratio bases right or everything downstream is wrong: DSO runs on revenue, DOH and DPO both run on COGS.
- Forgoing 2/10 net 30 is borrowing at a 44.6% EAR. Interest is the discount, the term is only the extra 20 days, and the base is 98, not 100.
- Rank liquidity current ≥ quick ≥ cash — and read them together. Licht's current ratio rose while it became less liquid, because the rise was inventory.
- NPV wins every disagreement. IRR is fine for a single conventional project and breaks on multiple roots, scale mismatch and its own reinvestment assumption.
- Three principles: after-tax cash flows, incremental only (sunk costs out, cannibalisation in), and correct timing. Then three cognitive errors and four behavioral biases.
- ROIC is the capital-efficiency test an outsider can compute — margin × turnover, compared against the required return, read as a trend not a single year.
- Real options are the right, not the obligation, to change course: Gerhardt goes from −90.91 to +8.26 once the facility's alternative uses are priced in.